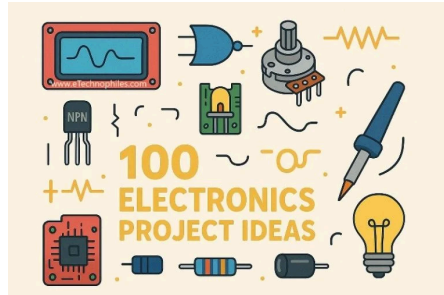


100 Electronics Projects Ideas For EE Students

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In this article, we've curated a diverse list of 100+ Electronics Projects, clearly categorized by difficulty: Beginner, Intermediate, and Advanced. Explore innovative ideas—from basic circuits and sensor applications to complex IoT devices and automation systems—that will inspire your next build.

Time to pick your project and level up your electronics skills!

Electronics Projects List

- [Beginner-Level Electronics Projects](#)
- [Intermediate Level Electronics Project](#)
- [Advanced Level Electronics Projects](#)

Beginner-Level Electronics Projects

1. LED Blinking Circuit using Arduino

A classic starter project for anyone new to electronics. You'll learn how to program an Arduino board to make an LED blink at specific intervals, mastering the basics of coding and circuits.

Start by connecting an LED and a resistor to an Arduino board. Upload a simple Arduino sketch to control the LED's blinking rate using digital output pins. This project teaches you basic Arduino programming, circuit connections, and digital signals.

2. Simple LED Flasher using 555 Timer IC

Create an LED flasher without programming using the versatile 555 Timer IC. This project helps you understand timer circuits and component interactions in analog electronics.

Build the circuit by connecting a 555 Timer IC in astable mode with resistors, capacitors, and an LED. Adjust the resistor and capacitor values to vary the flashing frequency.

3. Automatic Night Lamp using LDR and Transistor

Build a lamp that automatically turns on at night using basic components like an LDR and transistor, introducing fundamental analog electronics principles.

Connect an LDR in series with a resistor to create a voltage divider, feeding the transistor's base. As ambient light decreases, the transistor activates the LED or lamp connected to its output.

4. Temperature Monitor using Arduino and LM35

Interface the LM35 sensor with Arduino, write code to read temperature values, and display them on an LCD or serial monitor, strengthening your sensor-interfacing skills.

5. Water Level Indicator using Transistors

Design a straightforward circuit to indicate water levels in a tank. This project enhances your understanding of transistors and basic sensor mechanisms.

Insert metal probes at different heights in a water container and connect them to the transistor bases. When water touches the probes, transistors activate respective LEDs, indicating the water level visually.

6. Simplest weather station using DHT11 and Arduino

Learn how to accurately measure temperature and humidity using the DHT11 sensor combined with Arduino. This simple weather station displays real-time temperature plus humidity readings digitally, teaching you essential concepts of analog-to-digital conversions.

7. Light Sensor Alarm Circuit (LDR + Buzzer)

Create an effective alarm system that activates automatically in the dark using an LDR (light-dependent resistor). This easy-to-build circuit teaches fundamental sensor behavior and basic transistor switching techniques.

Assemble a simple circuit using an LDR, transistor, resistor, and a buzzer. When the ambient light decreases below a set threshold, the LDR's resistance triggers the transistor to activate the buzzer, creating an audible alert.

8. IR Remote Control for LEDs using Arduino

Control your LEDs remotely with a simple infrared remote control. This project introduces you to wireless remote control techniques, IR sensor interfacing, and basic Arduino programming.

Interface an IR receiver sensor with an Arduino, decode signals from an IR remote, and use these signals to switch LEDs on and off. The Arduino code handles incoming IR commands, allowing wireless control of your circuit.

9. Basic Logic Gates Circuit (AND, OR, NOT with LEDs)

Build simple logic gate circuits (AND, OR, NOT) visually demonstrated with LEDs. Ideal for beginners in digital electronics, this project provides hands-on experience with logic gate behavior and truth tables.

Use basic logic gate ICs (such as 7408, 7432, and 7404), switches, resistors, and LEDs on a breadboard. Set inputs manually with switches and observe the LED outputs to visually understand how each logic gate operates.

10. Mini Piano with Arduino and Piezo Buzzer

Create a simple musical instrument using Arduino and a piezo buzzer. You'll gain practical experience in Arduino programming, handling digital input, and audio output concepts.

Connect pushbuttons to Arduino's digital input pins and a piezo buzzer to a digital output pin. Write Arduino code that produces distinct tones when each button is pressed, creating a mini electronic piano.

11. Basic LED Chaser using CD4017 IC

Create a visually appealing LED chaser circuit using the CD4017 decade counter IC. Ideal for beginners, this project introduces you to IC-based sequential circuits, clock pulses, and basic electronic timing principles.

Connect the CD4017 IC with a 555 timer circuit configured as an astable oscillator. The 555 timer generates clock pulses, sequentially lighting up LEDs connected to the CD4017 outputs, creating a chasing effect.

12. Ultrasonic Distance Meter using Arduino

Design your own distance measurement tool using Arduino and an ultrasonic sensor (HC-SR04). Perfect for learning sensor interfacing, Arduino programming, and practical measurement applications.

Connect the ultrasonic sensor to Arduino, send ultrasonic pulses toward an object, and measure the time taken for waves to reflect back. Use this data to calculate and display distance measurements on an LCD or serial monitor.

13. Clap Switch using 555 Timer

Build a simple clap-activated switch without using any microcontroller. Learn how sound detection, transistor amplification, and 555 timer circuits work practically in everyday electronics.

Assemble a microphone, transistor amplifier stage, and a 555 timer configured as a bistable switch. When a clap is detected, the microphone picks up the sound, activating the transistor, which triggers the 555 timer to toggle an LED or relay.

14. Soil Moisture Detector with LED Indicator

Construct a straightforward soil moisture detection circuit that visually indicates the moisture content in soil. Ideal for beginners interested in gardening or agriculture-related electronics.

Use two conductive probes inserted into the soil connected to a transistor circuit. When soil moisture decreases below a certain threshold, conductivity changes, triggering a transistor to illuminate an LED indicator.

15. Digital Dice with Arduino and 7-Segment Display

Make your own digital dice for games using Arduino and a 7-segment display. Learn about random number generation, 7-segment display interfacing, and basic Arduino programming.

Wire a 7-segment display to Arduino digital pins, program the Arduino to generate random numbers between 1 and 6 upon pressing a pushbutton, and visually display the dice number instantly.

16. Touch Sensor Light using Arduino

Design an interactive lighting system controlled by a simple touch sensor module and Arduino. An exciting beginner project for exploring capacitive touch sensing technology and Arduino programming.

Connect a capacitive touch sensor module to Arduino digital inputs. Program Arduino to detect touch inputs, turning on or off an LED or relay-driven lamp accordingly.

17. DC Motor Speed Control using 555 Timer PWM

Control the speed of a DC motor precisely using pulse-width modulation (PWM) generated by a 555 timer IC. A valuable project to understand motor control techniques without microcontrollers.

Configure a 555 timer IC in astable mode to generate PWM pulses. Adjusting a potentiometer in the circuit varies the pulse width, allowing precise speed control of the connected DC motor.

18. Motion-Activated Alarm using PIR Sensor and Arduino

Develop a simple security alarm system triggered by human movement using Arduino and a PIR sensor. This practical project introduces motion detection, Arduino interfacing, and alarm triggering concepts.

Connect a PIR (Passive Infrared) motion sensor and a buzzer to Arduino. Program the Arduino to monitor PIR signals, activating the alarm (buzzer) when motion is detected.

19. FM Transmitter Circuit

Create your own basic FM transmitter circuit to broadcast audio signals over short distances. Perfect for beginners to explore radio frequency circuits, antenna basics, and analog electronics principles.

Use a transistor-based oscillator circuit, an electret microphone, and an antenna. When powered, the microphone audio signals modulate the RF oscillator frequency, transmitting audio signals to any nearby FM radio.

20 Simple Audio Amplifier using LM386 IC

Build a compact audio amplifier circuit using the popular LM386 amplifier IC. Ideal as a beginner project for

Wire the LM386 amplifier IC on a breadboard with capacitors, resistors, and a speaker. Feed audio signals (like from a smartphone) into the circuit, amplifying the sound clearly through the speaker

21. Servo Motor Control using Potentiometer and Arduino

Gain precise control over a servo motor using Arduino and a potentiometer. This project introduces beginners to the basics of servo control, analog inputs, and Arduino coding.

Connect a potentiometer to an Arduino analog input pin and a servo motor to a digital PWM pin. Write Arduino code to read the potentiometer's position, translating it into servo angles, allowing smooth motor rotation according to user input.

22. Burglar Alarm using Magnetic Reed Switches

Build a simple and effective burglar alarm using magnetic reed switches. Ideal for beginners interested in home security, you'll learn about switches, transistor circuits, and basic electronics triggering mechanisms.

Place magnetic reed switches at doors or windows connected to a transistor-driven buzzer circuit. When the

23. Adjustable Power Supply using LM317

Create your own adjustable power supply circuit using the versatile LM317 voltage regulator IC. Perfect for beginners interested in power electronics, voltage regulation, and practical circuit building.

Set up an LM317 regulator circuit on a breadboard with resistors and capacitors. Adjust the output voltage easily by varying the resistance of a connected potentiometer, giving you a flexible, adjustable voltage output suitable for powering various circuits.

24. RFID Door Lock using Arduino

Make a smart RFID-based door lock using Arduino. This exciting project introduces beginners to RFID technology, access control systems, and Arduino programming for security applications.

Interface an RFID reader module with Arduino. Program it to detect authorized RFID tags or cards, triggering Arduino-controlled relays or solenoids to lock and unlock doors securely.

25. Continuity Tester Circuit

Construct a handy continuity tester tool to check wires, connections, or circuit paths easily. An essential beginner-friendly project that helps in troubleshooting and understanding fundamental electronic circuits.

Set up a simple transistor-based circuit with an LED or buzzer on a breadboard. The LED or buzzer activates when continuity is detected, indicating a good connection or wire continuity during circuit testing.

26. IR Obstacle Detection Circuit (without microcontroller)

Create a simple obstacle detection circuit using IR sensors without any microcontroller. Ideal for beginners interested in sensors and analog electronics, this project teaches the basics of infrared sensing, transistor switching, and practical circuit design.

Build an IR LED transmitter and photodiode receiver circuit on a breadboard. When an obstacle approaches, IR rays reflect back onto the photodiode, triggering a transistor circuit that lights up an LED or activates a buzzer to indicate detection.

27. Digital Clock using Arduino and RTC Module (DS3231)

Build an accurate digital clock using an Arduino board and a Real-Time Clock (RTC) module like the DS3231. This project helps beginners explore I2C communication, accurate timekeeping, and Arduino programming concepts.

Connect a DS3231 RTC module to Arduino using I2C pins. Write Arduino code to read time data from the RTC module and display it clearly on an LCD or OLED screen, creating an easy-to-use digital clock with precise time accuracy.

28. Simple Relay Control Circuit Using Transistors

Learn to safely control high-voltage appliances using a low-voltage transistor-based relay circuit. This straightforward project introduces relay operation, transistor switching, and isolation principles.

Wire a transistor to a relay coil on a breadboard, controlled by a low-voltage input signal (like a switch or sensor). When triggered, the transistor activates the relay, safely switching on or off AC/DC appliances.

29. Morse Code Flasher using Arduino

Make your own Morse code flasher system using Arduino. An exciting beginner-friendly project introducing Arduino programming, timing sequences, and digital output concepts through practical Morse code signaling.

Connect an LED to an Arduino digital output pin. Write Arduino code to translate simple text into Morse code patterns, blinking the LED accordingly to visually display Morse messages.

30. Temperature Controlled Fan using Thermistor and Op-Amp

Design a practical fan-control circuit that automatically regulates fan speed based on temperature, without any microcontroller. A valuable project to understand analog temperature sensors, operational amplifiers, and transistor control circuits.

Use an NTC thermistor as a temperature sensor connected to an operational amplifier (e.g., LM358 or LM741) configured as a voltage comparator. The comparator output drives a transistor circuit, automatically switching on or adjusting the speed of a DC fan as temperature changes.

Intermediate Level Electronics Project

Build a robot that moves in response to your hand gestures. This project helps you dive deeper into robotics, wireless communication, and motion sensing.

Use an accelerometer sensor (like MPU6050) mounted on your hand to detect gestures. Send these signals wirelessly via RF or Bluetooth modules to an Arduino-based robot. The Arduino interprets signals, controlling DC motors to move the robot forward, backward, or sideways based on your gestures.

32. Fingerprint-Based Attendance System

Create an advanced attendance management system using fingerprint biometrics. This project introduces you to fingerprint sensors, databases, and biometric authentication techniques.

Interface a fingerprint sensor module with Arduino or ESP32. Enroll fingerprints into the sensor's memory, then verify attendance by matching scanned fingerprints against stored records. Display attendance status on an LCD or OLED screen, and optionally log the data for record-keeping.

33. Bluetooth-Controlled Home Automation

Connect a Bluetooth module (HC-05 or HC-06) to Arduino along with relays for appliance switching. Write Arduino code to receive commands from a smartphone app, allowing wireless control of lights, fans, or other appliances easily.

34. Wi-Fi Weather Station using ESP32 and OLED Display

Construct a portable weather station that measures temperature, humidity, atmospheric pressure, and displays data wirelessly on an OLED screen. Ideal for learning Wi-Fi connectivity, cloud integration, and advanced sensor interfacing.

Connect weather sensors (e.g., BME280) to an ESP32. Program the ESP32 to gather environmental data, send it wirelessly to cloud platforms or local web servers, and display the readings on an OLED display, providing accurate real-time weather data.

35. Automatic Solar Tracking System

Design a solar panel system that automatically follows the sun's movement for maximum efficiency. An excellent project introducing intermediate-level mechanical integration, sensors, and solar energy principles.

Use LDR sensors positioned around the solar panel to detect sunlight intensity. Arduino or a similar microcontroller reads sensor inputs and controls servo motors or DC motor-based actuators to adjust panel orientation, ensuring the panel faces the brightest sunlight continuously.

36. RFID-Based Attendance and Access Control System

Create an efficient RFID-based system for attendance logging and secured door access control. Learn RFID technology, database storage, access control, and integration with Arduino-based systems.

Interface an RFID reader with Arduino, store authorized RFID tags or card IDs in code or memory, and verify scanned IDs. Upon successful verification, Arduino can activate a relay or solenoid lock to grant access, and record attendance data digitally on an SD card or serial monitor.

37. Digital Oscilloscope using Arduino and OLED Display

Build a portable, cost-effective oscilloscope using Arduino and an OLED display. Ideal for intermediate-level electronics enthusiasts looking to learn about signal analysis, waveform visualization, and advanced Arduino programming.

Connect analog input signals to Arduino's analog pins and display real-time waveforms on an OLED screen. Arduino captures the input signal, converts analog signals to digital, and plots them graphically, allowing basic analysis of voltage signals and waveforms.

38. GSM-based Home Security Alarm System

Create a home security system that sends alerts to your phone via GSM when an intrusion is detected. This practical project teaches GSM module interfacing, sensor integration, and real-time alert systems.

Integrate sensors like PIR motion detectors or magnetic switches with Arduino and a GSM module (SIM800L). When a sensor triggers, Arduino sends an SMS or calls your phone, alerting you instantly of any security breach.

39. Voice-Controlled Robot using Arduino and Bluetooth

Develop a robot that moves according to voice commands from your smartphone. An exciting intermediate project that covers Bluetooth communication, robotics, and speech-controlled automation.

Pair Arduino with an HC-05 Bluetooth module and a motor driver circuit. Send voice commands through a dedicated smartphone app, convert them to commands via Bluetooth, and Arduino executes these commands to control robot movements.

40. GPS-Based Vehicle Tracking System

Design an accurate vehicle tracking system using GPS technology. Perfect for understanding GPS modules, data logging, and real-time location tracking techniques

Interface a GPS module (like NEO-6M) with Arduino to capture real-time coordinates. Arduino processes the GPS data and can log it on an SD card or transmit it through GSM/Wi-Fi modules, allowing continuous monitoring and tracking of vehicles.

41. Biometric Fingerprint Door Lock with Arduino

Upgrade your home or office security by creating a biometric fingerprint-based lock system. Explore

Connect a fingerprint sensor (R307, R305) to Arduino, enroll and verify fingerprints, and control a relay or solenoid lock based on successful authentication. Arduino programming handles fingerprint matching and provides user-friendly feedback through an LCD display.

42. Smart Blind Stick using Ultrasonic Sensor and Arduino

Build an intelligent blind stick to assist visually impaired individuals using ultrasonic sensors. Learn advanced sensor interfacing, haptic feedback, and practical applications of assistive technology.

Attach an ultrasonic sensor (HC-SR04) to an Arduino-powered stick. Arduino continuously measures distances to obstacles, alerting the user via vibrations from a connected vibration motor or sound from a buzzer, helping the user navigate safely.

43. Line Following Robot with PID Control

Enhance your robotics skills by building a sophisticated line-following robot using PID (Proportional-Integral-Derivative) control for smooth navigation. This project introduces intermediate robotics, feedback loops, and advanced Arduino programming techniques.

Equip your robot with IR sensors or photodiodes to detect line positions. Arduino continuously calculates the error between the sensor's input and desired path, adjusting motor speeds using PID algorithms, resulting in stable and accurate robot navigation.

44. Arduino-Based MP3 Player using DFPlayer Module

Build your own compact MP3 player using Arduino and a DFPlayer mini module. Ideal for exploring audio playback technology, serial communication, and Arduino audio applications.

Connect the DFPlayer Mini to Arduino and load audio files onto an SD card. Write Arduino code to select and play audio files, control volume, and handle user inputs via push buttons or an IR remote for easy audio playback control.

45. Obstacle Avoidance Drone using Arduino and Ultrasonic Sensors

Create a drone or quadcopter that automatically detects and avoids obstacles using ultrasonic sensors and Arduino. Great for intermediate-level students keen to explore drones, autonomous flight, and sensor integration.

Mount ultrasonic sensors (HC-SR04) on the drone's frame, interfaced with Arduino or flight controllers. Sensors detect nearby objects, prompting the drone to automatically adjust flight paths, ensuring safe autonomous flying.

46. Wi-Fi Controlled Robot using ESP8266/ESP32

Construct a robot you can remotely control from your smartphone or web browser over Wi-Fi. Perfect for diving deeper into IoT concepts, wireless communication, and web-based controls.

Use an ESP8266 or ESP32 board to build a Wi-Fi-enabled robot. Develop a simple web-based interface or smartphone app to send commands wirelessly to the robot, controlling motors for movement, direction, and speed in real-time.

47. Automatic Pet Feeder with Timer and Servo Motors

Design a smart automatic feeder that dispenses food to pets at preset intervals. Ideal for intermediate-level electronics enthusiasts interested in automation, servo motors, and practical home applications.

Set up a timer using an Arduino or RTC module (DS3231). At defined intervals, Arduino activates servo motors to open a feeding compartment, releasing a controlled portion of food automatically for your pet.

48. AC Appliance Control using TRIAC and Arduino

Safely and efficiently control AC appliances using Arduino and TRIAC circuits. Explore power electronics, AC voltage control, and isolation techniques.

Connect a TRIAC driver circuit (like MOC3021 optocoupler) to Arduino digital output pins. Arduino triggers the TRIAC circuit, providing smooth and safe switching for AC appliances such as lamps, fans, or heaters through controlled voltage adjustments.

49. Wireless Power Transfer Circuit

Build a simple wireless charging circuit that transmits power through inductive coupling. Perfect for intermediate learners curious about wireless energy transmission, electromagnetic fields, and inductors.

Construct a primary coil powered by a high-frequency oscillator circuit and a secondary coil with rectifier circuitry for power reception. Transmit power wirelessly through magnetic coupling, demonstrating inductive power transfer principles practically.

50. Portable Digital Multimeter using Arduino

Create your portable digital multimeter capable of measuring voltage, current, and resistance with Arduino. A valuable project that enhances understanding of instrumentation, measurement principles, and Arduino programming.

Interface voltage dividers, shunt resistors, and analog-to-digital converters (built into Arduino) to measure electrical parameters. Display real-time readings clearly on an LCD or OLED, creating your personalized portable testing device.

51. Heart Rate Monitoring System using Pulse Sensor

Build your own portable heart rate monitor using Arduino and a pulse sensor. Ideal for exploring biomedical sensors, analog-to-digital signal conversion, and real-time health monitoring applications.

display, providing immediate feedback.

52. Bidirectional Visitor Counter using Arduino and IR Sensors

Develop a smart system to count the number of visitors entering and exiting a room automatically. This intermediate project introduces infrared sensors, data handling, and practical automation techniques.

Set up IR sensors in pairs at entry/exit points connected to an Arduino. When someone passes through sensors, Arduino increments or decrements the count, displaying real-time visitor numbers on an LCD or serial monitor.

53. Bluetooth-Controlled Smart Car with Smartphone App

Construct an advanced Bluetooth-enabled robotic car controllable directly via your smartphone. A fun, interactive project to learn robotics, Bluetooth communication, and mobile app integration.

Combine an Arduino board with an HC-05 Bluetooth module and motor driver (like L298N). Develop or use a smartphone app to wirelessly send directional commands, allowing smooth remote control over your robotic vehicle's movements.

54. Soil Monitoring IoT System using ESP32 and Cloud

Create an IoT-based soil monitoring system for agriculture or gardening, sending real-time data to the cloud. Ideal for learning IoT integration, sensor interfacing, and remote data visualization.

Connect soil moisture and temperature sensors to ESP32. Program the ESP32 to upload sensor data periodically to cloud platforms like ThingSpeak or Arduino IoT Cloud. Monitor and analyze soil conditions remotely from anywhere via smartphone or web.

55. Arduino-Based Spectrum Analyzer

Make your audio spectrum analyzer using Arduino, visualizing frequency bands of audio signals. A fascinating intermediate-level project for those interested in audio processing, signal visualization, and advanced Arduino programming.

Interface a microphone or audio input with Arduino's analog input. Arduino performs FFT (Fast Fourier Transform) calculations on incoming audio signals, visually displaying frequency bands and intensity on LEDs, LCD, or OLED displays.

56. Automatic Railway Crossing Gate Controller

Build an automated railway gate control system that safely manages gate openings based on train proximity. A practical project introducing automation, sensor integration, and safety systems.

Place IR or ultrasonic sensors near railway tracks connected to an Arduino. Sensors detect incoming trains, triggering Arduino to control servo motors or DC motors to close/open railway crossing gates automatically, ensuring passenger safety.

57. Automatic Room Occupancy Detector

Design a smart room occupancy detection system that automatically controls lighting and ventilation based on human presence. Perfect for understanding PIR sensors, automation logic, and relay interfacing.

Use PIR sensors connected to Arduino to detect human presence in a room. Arduino then controls relays connected to lights or fans, switching appliances automatically when occupants enter or leave, saving energy efficiently.

58. LED Cube Display (4x4x4)

Construct a visually captivating 3D LED cube to display animations and patterns. Ideal for exploring multiplexing, digital logic, soldering skills, and creative Arduino programming.

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Arrange 64 LEDs in a 4x4x4 cube matrix controlled by Arduino via multiplexing techniques. Program Arduino to light up LEDs sequentially, creating impressive visual effects and animated patterns.

59. Solar-Powered Charger with Battery Management

Develop a solar-powered battery charger circuit with effective battery management using Arduino. This practical intermediate project helps you understand renewable energy systems, battery charging methods, and energy management principles.

Interface solar panels with Arduino-controlled charging circuits (voltage/current regulators). Arduino monitors battery voltage and controls the charging process, ensuring safe and efficient energy storage in

60. Arduino Radar System using Servo and Ultrasonic Sensor

Build your own radar system that detects and visually displays object positions using ultrasonic sensors and servo motors. An exciting intermediate project to explore sensor positioning, data visualization, and real-time detection.

Mount an ultrasonic sensor (HC-SR04) onto a servo motor controlled by Arduino. Sweep the servo motor back and forth, scanning the environment. Arduino captures distance readings and visually displays the radar-like data on your computer screen or LCD.

Advanced level Electronics Projects

61. Autonomous Drone with GPS Navigation and Obstacle Avoidance

Create a sophisticated autonomous drone capable of navigating via GPS and intelligently avoiding obstacles. This advanced project combines drone flight mechanics, GPS interfacing, and real-time sensor fusion techniques.

continuously detect obstacles, prompting the drone to adjust its flight path autonomously for safe navigation.

62. Advanced Home Automation System using IoT and Cloud Integration

Build an advanced IoT-based smart home system to remotely control and monitor appliances from anywhere. Dive deep into wireless connectivity, cloud platforms, mobile app integration, and real-time data analytics.

Use ESP32 or Raspberry Pi boards to connect home appliances via Wi-Fi. Integrate sensors, relays, and cloud services (like AWS IoT, Blynk, or Home Assistant) for monitoring and remote control, enabling automation, scheduling, and intelligent decision-making based on real-time sensor data.

63. Facial Recognition Door Lock System using Raspberry Pi and OpenCV

Enhance security with an advanced door lock that uses facial recognition technology. Perfect for understanding computer vision, Raspberry Pi integration, machine learning, and security applications.

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transistor	>
amplifier circuit	>
transistors	>

verification, Raspberry Pi triggers an electronic door lock mechanism, ensuring secure and convenient access control.

64. Self-Balancing Robot using PID Control and IMU Sensor

Design a sophisticated two-wheeled robot capable of balancing autonomously. Ideal for advanced robotics enthusiasts interested in sensor integration, PID algorithms, and precise real-time control systems.

Use an IMU sensor (e.g., MPU6050) to measure tilt angles precisely. Implement PID (Proportional-Integral-Derivative) control algorithms on Arduino or ESP32 to drive motors and adjust the robot's balance dynamically, maintaining stable, upright orientation.

65. Smart Energy Meter using IoT (ESP32 and Cloud Data Logging)

Develop an IoT-enabled energy meter capable of real-time energy consumption monitoring and cloud-based analytics. Excellent for exploring IoT integration, power measurement, and data visualization.

Measure AC energy usage using specialized sensors (like ACS712 or PZEM modules) interfaced with ESP32. Send data wirelessly to cloud platforms (like ThingSpeak, Firebase, or Arduino IoT Cloud) for analysis, providing insights, real-time monitoring, and efficient energy management.

66. Speech Recognition Controlled Home Appliances using Raspberry Pi

Construct a smart home system controlled entirely by voice commands. This advanced project combines speech recognition, natural language processing, Raspberry Pi programming, and appliance automation.

Utilize Raspberry Pi integrated with microphones and voice-recognition libraries (Google Assistant SDK, Amazon Alexa, or offline solutions like PocketSphinx). Raspberry Pi interprets voice commands and controls relays connected to home appliances, enabling hands-free operation and intelligent home automation.

67. 3D Printer Controller Board (Custom PCB with Arduino/Raspberry Pi)

Design and build your own customized 3D printer control board, enhancing your skills in PCB designing, embedded systems, and precise motor control.

Create a custom PCB incorporating stepper motor drivers (like A4988 or DRV8825), microcontrollers (Arduino Mega/Raspberry Pi), temperature sensors, and LCD interfaces. Program firmware (Marlin or custom-written) to precisely control motors, heaters, and sensors for accurate 3D printing functionality.

68. Wireless Sensor Network for Agriculture Monitoring (Mesh Network)

Develop a wireless mesh network system for smart agriculture to monitor soil health, temperature, humidity, and crop conditions. Perfect for exploring wireless networking, IoT sensor nodes, and large-scale data collection.

Build multiple sensor nodes using ESP32 or ZigBee modules, each equipped with environmental sensors. Nodes communicate in a mesh topology, sending sensor data wirelessly to a central coordinator. The data is logged, analyzed, and visualized in the cloud, allowing efficient agricultural management.

69. IoT-based Patient Health Monitoring System with Cloud Analytics

Construct a sophisticated IoT-based healthcare system capable of remotely monitoring patient vital signs in real-time and providing analytics through cloud integration.

Integrate biomedical sensors (ECG, heart rate, temperature, oxygen saturation) with ESP32 or Raspberry Pi. Send sensor data securely to cloud services (AWS IoT, ThingSpeak, or Firebase) for remote access, monitoring, and real-time health alerts, enabling timely healthcare interventions.

70. CNC Machine Controller using Arduino or ESP32

Build a custom CNC (Computer Numerical Control) controller to precisely operate CNC machines such as engravers, routers, or milling machines. Ideal for mastering industrial automation, precise motor control, and real-time firmware development.

Develop firmware for Arduino or ESP32 integrated with stepper motor drivers and sensors. Interface with CNC software (like GRBL) to interpret G-code commands, providing accurate, programmable movements for CNC applications.

71. Advanced Weather Station with Predictive Analysis using Machine Learning

Design an intelligent weather station that not only measures current environmental parameters but also predicts future weather patterns using machine learning algorithms.

Integrate weather sensors (temperature, humidity, atmospheric pressure, wind speed) with Raspberry Pi or ESP32. Transmit collected data to a cloud or local server, implementing machine learning algorithms (linear regression or neural networks) to analyze trends and forecast weather conditions accurately.

72. Image Processing Robot for Color/Object Detection and Sorting

Develop an intelligent robot capable of detecting, identifying, and sorting objects based on color or shape using advanced image processing techniques.

Use a Raspberry Pi equipped with a camera module running OpenCV. Capture images in real-time, apply image processing algorithms to detect color or shape, and control actuators or robotic arms to sort objects automatically, teaching advanced computer vision, robotic control, and real-time decision-making.

73. Li-Fi Communication System (Wireless Data Transmission through Light)

Create a cutting-edge wireless communication system using visible light (Li-Fi), capable of transmitting high-speed data. Ideal for advanced learners interested in optical communication, modulation techniques, and innovative wireless technologies.

74. Electric Vehicle (EV) Battery Management System

Build a robust battery management system (BMS) designed specifically for electric vehicles. This project explores battery balancing, state-of-charge estimation, and advanced power electronics management.

Use microcontrollers (Arduino, STM32, or ESP32) and specialized battery-management ICs to monitor battery voltage, current, and temperature. Implement balancing algorithms to ensure optimal battery health and maximize performance.

75. Real-Time Vehicle Accident Detection and Alert System using IoT

Develop a system to detect vehicle accidents instantly and send real-time alerts to emergency contacts and services. Ideal for exploring accelerometer sensors, GPS modules, and IoT-based alert systems.

Integrate accelerometers, GPS modules, and GSM/Wi-Fi modules with Arduino or ESP32. The system detects sudden impacts or unusual movements, immediately sending location data and emergency alerts to designated contacts, potentially saving lives.

Design an intelligent wheelchair controlled through simple hand gestures. This project involves sensor integration, robotics, real-time gesture interpretation, and practical assistive technology applications.

Equip the wheelchair with gesture sensors (accelerometer/gyroscope modules or camera-based gesture detection). A microcontroller (like Arduino or Raspberry Pi) interprets gestures, controlling motor drivers for smooth, intuitive wheelchair navigation.

77. Smart Helmet for Accident Prevention using IoT and Sensors

Create a smart helmet designed to enhance rider safety by preventing accidents through IoT integration and real-time sensor monitoring.

Integrate impact sensors, accelerometers, GPS, and GSM modules with Arduino or ESP32. The helmet monitors critical parameters (impact detection, rider behavior, speed), providing real-time alerts to users or emergency services in case of accidents or unsafe riding conditions.

78. Automatic License Plate Recognition System using Raspberry Pi

Develop an advanced image-processing system to automatically identify vehicle license plates, useful for security, traffic monitoring, or automated parking solutions.

Combine a Raspberry Pi, camera module, and OpenCV software. Capture images in real-time, apply OCR (Optical Character Recognition) algorithms, identify license plate numbers, and log or transmit data for verification and analysis.

79. Solar-Powered Smart Irrigation System with Machine Learning

Construct an advanced solar-powered irrigation system capable of intelligently managing water usage through machine learning and environmental sensor data.

Integrate soil moisture sensors, weather sensors, solar panels, and water pumps controlled by ESP32 or Raspberry Pi. Collect data and use machine learning algorithms to predict optimal watering schedules, maximizing efficiency and sustainability.

80. Wireless Industrial Automation and Control System (SCADA)

Create a wireless Supervisory Control and Data Acquisition (SCADA) system suitable for industrial automation and monitoring. Explore advanced wireless communication, IoT-based data logging, and remote process control techniques.

Use wireless sensor nodes (ESP32, ZigBee, or LoRa modules) for data collection and actuator control. Centralized software monitors real-time sensor data, controls processes remotely, and ensures efficient operation of industrial equipment.

81. Voice-Controlled Drone with Advanced Flight Stabilization

Integrate voice recognition modules (using Raspberry Pi or ESP32) with a flight controller. Commands from voice inputs control drone navigation, while IMU sensors and PID controllers provide smooth stabilization and accurate flight handling.

82. FPGA-Based Digital Oscilloscope

Construct your high-performance oscilloscope utilizing FPGA technology. Ideal for mastering FPGA programming, signal processing, and real-time waveform analysis.

Program an FPGA board to capture high-speed analog signals using ADC interfaces, displaying signals graphically on an LCD or OLED. Implement FPGA-based algorithms to process signals quickly, accurately visualizing waveform data and measurements.

83. Multi-Robot Communication System using ZigBee and Swarm Robotics

Design a robust multi-robot communication network using ZigBee technology, ideal for swarm robotics and cooperative robotics applications.

Integrate multiple robots with ZigBee modules for secure, reliable communication. Program coordinated actions and data exchange among robots, enabling them to collaboratively perform tasks such as area exploration, object manipulation, or synchronized movements.

84. IoT-Based Air Quality Monitoring System with Data Visualization

Create a sophisticated air quality monitoring system capable of real-time measurement and cloud-based visualization. Perfect for advanced learners focusing on environmental monitoring, IoT systems, and data analytics.

Connect air quality sensors (e.g., MQ-series gas sensors, particulate matter sensors) to an ESP32 or Raspberry Pi. Transmit data to cloud platforms for real-time visualization and analytics, providing accurate insights into air pollution levels and environmental conditions.

85. Brainwave-Controlled Robotics using EEG Sensors

Explore cutting-edge brain-computer interface (BCI) technology by building a robot controlled entirely by brainwave signals.

Integrate EEG sensor headsets (such as Neurosky Mindwave) with Arduino or Raspberry Pi. Interpret brainwave patterns into actionable commands, allowing real-time robotic movements based on cognitive or focused signals, introducing users to advanced neuroscience and robotics interaction.

86. Real-Time Sign Language Recognition System

Develop an advanced image-processing system to recognize and translate sign language gestures in real-time. Ideal for understanding advanced computer vision, machine learning, and accessibility technology.

87. Smart Traffic Management System using IoT and AI

Design a smart traffic management system that optimizes traffic flow, reduces congestion, and improves safety using IoT sensors and artificial intelligence.

Deploy cameras, ultrasonic, or IR sensors connected to Raspberry Pi or ESP32 devices at traffic junctions. Implement AI algorithms to analyze real-time data, dynamically control traffic lights, and adapt traffic patterns, significantly improving transportation efficiency.

88. Automatic Autonomous Parking System

Construct an autonomous vehicle parking system capable of precise vehicle positioning and automated parking maneuvers using sensor integration and real-time feedback control.

Use ultrasonic sensors, LiDAR, or cameras interfaced with Arduino/Raspberry Pi. Algorithms process sensor inputs, calculating precise paths, steering, and motor actions, enabling vehicles to park autonomously into designated spots without human intervention.

89. Virtual Reality Glove for Motion and Gesture Tracking

Create an advanced virtual reality (VR) glove capable of tracking hand movements and gestures in real-time. Explore sensor fusion, gesture recognition, and immersive VR interaction technologies.

Embed flex sensors, accelerometers, and gyroscopes in a glove interfaced with Arduino or ESP32. Transmit sensor data wirelessly to VR environments, translating precise hand and finger movements into interactive virtual experiences.

90. Long-Range LoRa-Based Communication System

Design a robust, long-range wireless communication system using LoRa technology, perfect for remote monitoring, IoT deployment, and advanced data communication applications.

Build sender and receiver nodes using LoRa modules connected to ESP32 or Arduino. Achieve reliable data transmission over kilometers, ideal for remote sensor networks, environmental monitoring, agriculture applications, or IoT deployments requiring extensive coverage.



Ankit Negi

I am an electrical engineer by profession who loves to tinker with electronic devices and gadgets and have been doing so for over six years now. During this period, I have made many projects and helped thousands of students through my blog and YouTube videos. I am active on [LinkedIn](#).

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